

HBC RADIO

ATAK Plugin — Interface Control Document

Ham Binary CoT over Distance-Vector Mesh Radio

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1. Overview

HBC Radio is an ATAK plugin that transmits Cursor-on-Target (CoT) situational-awareness data between EUDs over amateur-radio audio links, with no network infrastructure. The stack has three interfaces, each specified in this document:

- **Physical layer** — one of three selectable audio modems (AFSK1200, OFDM COFDMTV, or Mercury HF / FreeDV DATAC4) carrying opaque byte frames over a voice radio.
- **Mesh network layer** — an uncoordinated distance-vector routing protocol (announce propagation, hop-by-hop directs, end-to-end ACKs) adapted from the Reticulum routing model for unencrypted FCC Part 97 operation.
- **Application layer** — the HBC Protocol v1.6, a compact bit-packed binary encoding of CoT events (PLI, markers, chat, alerts, shapes, CASEVAC).

NOTE: All content is transmitted in cleartext. Callsigns ride in every frame header, satisfying station-identification requirements.

2. System Architecture

Data path for one transmission:

ATAK CoT event → HBC encoder (binary payload) → MeshRouter (13+ byte header, routing) → modem framing → audio → radio (VOX or PTT)

Reception is the mirror image. The mesh layer decides whether a received frame is delivered locally, forwarded toward its destination, rebroadcast (announces), or dropped.

Layer	Implementation	Addressing
Application	HbcEncoder / HbcDecoder	Callsign in HBC header; deterministic UUIDs
Network	MeshRouter	Origin / Transmitter / Destination / Next-hop callsigns (ITA2)
Physical	AudioModem, OfdmModem, MercuryModem	None (byte pipe; AFSK carries fixed AX.25 wrapper for station ID)

3. Physical Layer (Audio Modems)

All three modems carry the identical mesh frame; the modem must match on all stations sharing a channel.

Parameter	AFSK1200	OFDM COFDMTV	Mercury HF
Waveform	Bell 202 AFSK, 1200 baud	QPSK OFDM (rattlegram "short"), 1600 Hz BW	FreeDV DATAC4 raw-data OFDM (codec2)
Sample rate	48000 Hz	48000 Hz	8000 Hz
Payload capacity per burst	Practical AX.25 limits (>200 B)	169 B	52 B (54 B frame minus framing)
FEC / integrity	HDLC CRC-16	Polar code + CRC-32	LDPC + CRC-16
Intended use	VHF/UHF FM, packet compatible	VHF/UHF FM, robust to device DSP	HF SSB, very low SNR

3.1 AFSK1200 / AX.25

Mesh frames are carried as the information field of a single AX.25 UI frame. The AX.25 addresses are fixed PHY framing only and are not used for routing: destination HBC, source = operator ham callsign (Part 97 ID), no digipeater path. TX renders the entire burst in advance: lead silence (150 ms), optional VOX leader tone, HDLC flag preamble ("TX Dwell", default 500 ms), frame, tail silence (250 ms).

3.2 OFDM COFDMTV (rattlegram)

Payload framing inside the 170-byte OFDM block: byte 0 = mesh frame length (1–169), bytes 1..n = mesh frame, zero-padded. The operator callsign also rides in the OFDM metadata header. Carrier 1500 Hz; VOX leader implemented as noise symbols (~180 ms each).

3.3 Mercury HF (FreeDV DATAC4)

Burst layout matches the Mercury HF modem exactly: preamble → one DATAC4 frame with CRC-16 in the final two bytes → postamble, with 100 ms head/tail silence. Payload framing inside the 54-byte usable frame: byte 0 = length, byte 1 = 'H' domain marker, bytes 2..n = mesh frame, zero-padded.

NOTE: This interface speaks Mercury's waveform/frame/CRC layout, not Mercury's ARQ data-link protocol. Desktop Mercury demodulates these frames but does not route them to its TCP interface.

3.4 Channel Access (CSMA)

Every modem performs carrier-sense multiple access before transmitting: RX audio energy is tracked against an adaptive noise floor (busy above 4× floor, held 400 ms past the last loud chunk). Before TX the station waits for a clear channel plus a random 150–550 ms backoff that

must remain clear; after 8 s (15 s on Mercury) the frame is sent regardless so traffic cannot be starved. AFSK additionally uses the demodulator's DCD as a second carrier-sense input.

4. Mesh Network Layer

Implements “Adaptation of Uncoordinated Distance-Vector Routing for Unencrypted Amateur Radio Networks” (see References), with one approved deviation: packet payloads carry HBC binary rather than ITA2 text.

4.1 Common Packet Header (13 bytes)

Byte offset	Length	Field	Description
0	1	Packet Type	0x01 Announce, 0x02 Broadcast, 0x03 Direct, 0x04 ACK
1	5	Origin Callsign	Bit-packed ITA2 (section 4.2). The station that created the packet.
6	5	Transmitter Callsign	The station transmitting this specific hop; rewritten by relays.
11	2	Sequence ID	Unsigned 16-bit, big-endian. Loop prevention (section 4.8).

4.2 Callsign Encoding (ITA2, 5 bytes)

Callsign fields are exactly 8 five-bit ITA2 codes bit-packed MSB-first into 5 bytes (40 bits), NULL (00000) padded. Shifts: LTRS = 11111, FIGS = 11011. A leading LTRS is emitted before the first letter; when a callsign would need 9 codes (e.g. KE8TQB), the leading LTRS is omitted — decoders start in letters state, so both forms decode identically.

Example	Code sequence	Packed bytes (hex)
K1ABC	LTRS K FIGS 1 LTRS A B C	FF B7 DF E2 6E
N3DEF	LTRS N FIGS 3 LTRS D E F	F9 B7 0F CA 16
W2XYZ	LTRS W FIGS 2 LTRS X Y Z	FE 77 9F DE B1

NOTE: The source paper’s K1ABC example lists byte 03 as BF, but its own bit-level breakdown evaluates to DF; this implementation follows the arithmetic. All vectors above are enforced by the automated test suite (codec-test/MeshRouterTest.java).

4.3 Packet Type 0x01 — Announce (14 bytes)

Byte offset	Length	Field	Description
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0–12	13	Common Header	Type = 0x01
13	1	Hop Count	Unsigned 8-bit; origin sets 0, each relay increments

4.4 Packet Type 0x02 — Broadcast (13 + n bytes)

Byte offset	Length	Field	Description
0–12	13	Common Header	Type = 0x02
13+	variable	Payload	HBC binary message (section 5). Unrouted, unacknowledged.

4.5 Packet Type 0x03 — Direct Message (23 + n bytes)

Byte offset	Length	Field	Description
0–12	13	Common Header	Type = 0x03
13	5	Destination Callsign	Final recipient (ITA2)
18	5	Next Hop Callsign	Neighbor receiving this hop (ITA2); rewritten by each relay
23+	variable	Payload	HBC binary message

A station whose callsign matches Next Hop looks the destination up in its routing table, rewrites Next Hop and Transmitter, and re-transmits. The final destination delivers the payload and returns an ACK. With no route, the packet is dropped (relays) or sent as a Broadcast fallback (originator).

4.6 Packet Type 0x04 — Acknowledgment (20 bytes)

Byte offset	Length	Field	Description
0–12	13	Common Header	Type = 0x04. Origin = acknowledging station.
13	5	Destination Callsign	Origin of the Direct being acknowledged
18	2	Acked Sequence ID	Sequence ID of the confirmed Direct (big-endian)

Intermediate stations forward ACKs toward the destination when a route is known; the dedup cache prevents forwarding storms.

4.7 Route Discovery and Maintenance

- **Announce propagation** — unseen announces are rebroadcast with hops+1 and the local callsign as Transmitter; receivers record Destination = Origin, Next hop = Transmitter. Lowest hop count wins; equal-cost ties resolve “first heard.”
- **Passive learning** — hearing *any* valid frame registers its Transmitter as a 0-hop neighbor and (for non-announce frames) the Origin as reachable via that Transmitter. A single received PLI is sufficient to make a station addressable.
- **Announce-on-traffic** — a station’s own transmissions serve as its announce; an explicit Announce is sent only after 10 minutes with no transmissions (keepalive).
- **Expiration** — routing entries expire 30 minutes after the last frame that refreshed them; cleanup runs every 60 s.

4.8 Loop Prevention

Receivers cache [Origin + Sequence ID + Type] signatures for 5 minutes. A packet whose signature is cached is silently discarded (except a duplicate Direct addressed to this station, which is re-ACKed without re-delivery to survive a lost ACK).

4.9 Reliability and Retries

Directs operate fire-and-wait: the originator starts a retry timer of 5 s plus 0–2 s random jitter. If no ACK arrives, the Direct is re-transmitted with the **same** Sequence ID, up to 3 retries, after which the transmission is declared failed (logged). Broadcasts are never acknowledged or retried.

5. Application Layer (HBC Protocol v1.6)

The payload of every Broadcast/Direct is one HBC message: a bit-packed binary encoding of a CoT event. Full field-level definitions and worked byte examples are maintained in the HBC-Protocol repository (see References); this section summarizes the interface.

5.1 HBC Header

Field	Size	Description
Callsign	variable (ITA2, CR-terminated)	Sending operator/device callsign, max 8 characters
Version	3 bits	000 = version 1
Mode	3 bits	mode-1 for Modes 1–6; 111 = Mode 0 Ack

NOTE: HBC uses its own ITA2 table (LSB-first relative to the mesh header table) with CR terminators — the two layers' callsign encodings are related but not interchangeable.

5.2 Message Modes and Sizes

Mode	CoT type(s)	Content	Typical size
0 Ack	b-t-f-d / b-t-f-r	DM delivered/read receipt: recipient callsign, kind (2 b), 16-bit tag	~12 B
1 PLI / Spot	a-f-G, a-h-G, a-n-G, a-u-G	Spot flag, affiliation (2 b), name (7 ch), lat/lon (21+22 b)	~15 B
2 Alert	b-a-o-tbl / b-a-o-can	Active flag, alert + originator names, position	~18 B
3 GeoChat	b-t-f	Destination kind (2 b: All / Room / DM), room or recipient, DM 16-bit tag, ITA2 text	DM “helo landing zone” = 23 B
4 Shape	u-d-c-c / u-d-r / u-d-f	Circle (center + radius) or polygon/polyline (≤15 points)	varies
5 CASEVAC	b-r-f-h-c	MEDEVAC 9-line summary fields	~20 B
6 Extended Marker	any placed marker	Full CoT type token, 2525C/spot-map/custom icon reference, color tint	varies

5.3 Deterministic UUIDs

Object	UID rule
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PLI marker	HBC-{CALLSIGN}
911 alert	HBC-{CALLSIGN}-911
DM conversation	Recipient's UID; the receiving plugin substitutes its real ATAK device UID so the message files into the chat window
Spots / shapes / markers	Random UUID per injection

5.4 Direct-Message Tags and Receipts

Every chat DM carries a 16-bit tag = CRC-16/CCITT-FALSE (poly 0x1021, init 0xFFFF) of the ATAK `messageId`. The receiving EUD's automatic delivered/read receipts are converted to Mode 0 Acks echoing that tag; the sender maps the tag back to the original `messageId` and injects the receipt CoT, producing ATAK's delivered/read checkmarks end-to-end over RF. These application receipts are independent of the mesh-layer ACKs of section 4.6.

5.5 Contact Publication

Received PLIs are reconstructed with `<contact callsign="..." endpoint="*:-1:stcp"/>` and a `<__group>` detail, so ATAK registers the heard station as a messageable contact (chat DM list, send-to pickers) from a single PLI. Outgoing traffic to that contact is intercepted by the plugin's `PreSendProcessor` and transmitted over the mesh.

6. Timing Parameters and Constants

Parameter	Value	Layer
Route entry TTL	30 min (refreshed by any heard frame)	Mesh
Dedup cache TTL	5 min	Mesh
Cleanup interval	60 s	Mesh
Direct retry timer	5 s + 0–2 s jitter, max 3 retries	Mesh
Quiet-time announce keepalive	10 min without own traffic	Mesh
CSMA busy threshold / hold	4× adaptive noise floor / 400 ms	PHY
CSMA backoff / give-up	150–550 ms clear / 8 s (15 s Mercury)	PHY
PLI rate limit	User setting, default 60 s	Application
RX duplicate window (frame hash)	30 s	Plugin

7. Compatibility and Constraints

- All stations on a channel must run the same plugin wire version (0.12+ mesh framing) and the same modem selection.
- Mercury DATAC4 limits routed Directs to ~29-byte HBC payloads (Broadcasts ~39 B). All standard messages fit; very large Mode 6 markers may require Broadcast or another modem.
- Callsigns are limited to 8 ITA2 characters (A–Z, 0–9, hyphen).
- No encryption is used or permitted; do not transmit sensitive data.

8. References

1. HBC Protocol reference implementation and specification — github.com/ke8tqb/HBC-Protocol
2. HBC ATAK Plugin source — github.com/ke8tqb/HBC-ATAK-Plugin
3. *Adaptation of Uncoordinated Distance-Vector Routing for Unencrypted Amateur Radio Networks* (routing protocol paper)
4. Mercury HF modem — mercury.hermes.radio; FreeDV/codec2 — github.com/drowe67/codec2
5. rattlegram / COFDM TV — github.com/aicodix/rattlegram